The total leakage current of all the electrical equipment in the installation must be taken into account. You find the leakage current of the motor in the sections on leakage current and leakage current (AC).

This product can cause a direct current in the protective earth conductor.

Overvoltage and undervoltage protection

Overvoltage and undervoltage may occur in case of unstable power supply or faulty installation. The motor stops if the voltage falls outside the permissible voltage range. The motor restarts automatically when the voltage is again within the permissible voltage range. Therefore, no additional protection relay is required.

The motor is protected against transients from the power supply according to EN 61800-3. In areas with high lightning intensity, we recommend external lightning protection.

Overload protection

If the upper load limit is exceeded, the motor automatically compensates for it by reducing the speed and stops if the overload condition persists.

The motor remains stopped for a set period. After this period, the motor automatically attempts to restart. The overload protection prevents damage to the motor. Consequently, no additional motor protection is required.

Overtemperature protection

The electronic unit has a built-in temperature sensor as an additional protection. When the temperature rises above a certain level, the motor automatically compensates for it by reducing the speed and stops if the temperature keeps rising. The motor remains stopped for a set period. After this period, the motor automatically attempts to restart.

Protection against phase unbalance

Three-phase motors must be connected to a power supply with a quality corresponding to IEC 60146-1-1, class C, to ensure correct motor operation at phase unbalance. This also ensures long life of the components.

Related information

9.2.1.2 Leakage current

9.2.2.2 Leakage current (AC)

5.5 Connection terminals

The descriptions and terminal overviews in this section apply to both single- and three-phase motors.

For maximum torques, see the section on other technical data.

Related information

9.4 Other technical data

5.5.1 Connection terminals, CRE, CRIE, CRNE, SPKE and MTRE pumps

The CRE, CRIE, CRNE, SPKE and MTRE pumps have a number of inputs and outputs enabling the pumps to be used in advanced applications where many inputs and outputs are required.

The pumps have the following connections:

- three analog inputs
- one analog output
- two dedicated digital inputs
- two configurable digital inputs or open-collector outputs
- Grundfos Digital Sensor input and output
- two Pt100/1000 inputs
- two LiqTec sensor inputs
- two signal relay outputs
- GENIbus connection.

See the figure below.

Digital input 1 is factory-set to be start-stop input where open circuit results in stop. A jumper is factory-fitted between terminals 2 and 6. Remove the jumper if digital input 1 is to be used as external start-stop or any other external function.

DANGER
Electric shock
Death or serious personal injury

- Make sure that the wires to be connected to the connection groups below are separated from each other by reinforced insulation in their entire lengths.

- Inputs and outputs

All inputs and outputs are internally separated from the mains-conducting parts by reinforced insulation, and galvanically separated from other circuits. All control terminals are supplied with protective extra-low voltage (PELV), thus ensuring protection against electric shock.

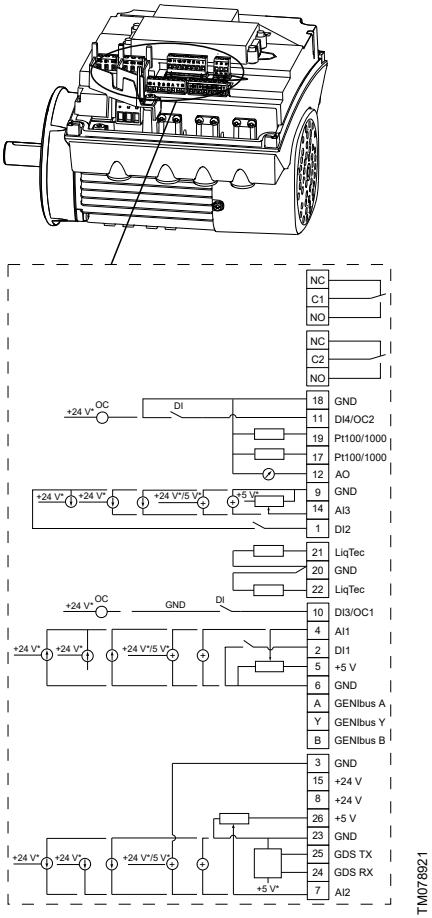
- Signal relay outputs
  - Signal relay 1:

    LIVE:

    You can connect supply voltages up to 250 VAC. PELV: The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or protective extra-low voltage to the output as desired.
  - Signal relay 2:

    PELV:

    The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or protective extra-low voltage to the output as desired.
- Mains supply (terminals N, PE, L or L1, L2, L3, PE).



Connection terminals, CRE, CRIE, CRNE, SPKE and MTRE pumps

*If you use an external supply source, there must be a connection to GND.

Terminal	Type	Function
2	DI1	Digital input, configurable
5	+5 V	Supply to potentiometer and sensor
6	GND	Ground
A	GENIbus, A	GENIbus, A (+)
Y	GENIbus, Y	GENIbus, GND
B	GENIbus, B	GENIbus, B (-)
3	GND	Ground
15	+24 V	Supply
8	+24 V	Supply
26	+5 V	Supply to potentiometer and sensor
23	GND	Ground
25	GDS TX	Grundfos Digital Sensor output
24	GDS RX	Grundfos Digital Sensor input
7	AI2	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V

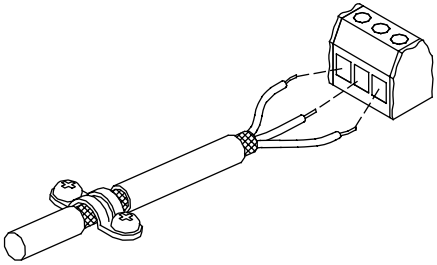
5.6 Signal cables

Use screened cables with a cross-sectional area of minimum 0.5 mm^2 and maximum 1.5 mm^2 for the external on/off switch, digital inputs, setpoint and sensor signals.

The wires in the motor terminal box must be as short as possible.

5.6.1 Connecting signal cables

1. Connect the screens of the cables to the frame at both ends with good connection. The screens must be as close as possible to the terminals.



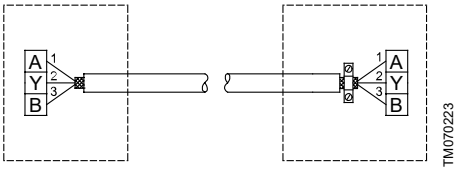
2. Connect the signal cables to the terminals.
3. Tighten the terminal screws.

5.7 Bus connection cable

5.7.1 Connecting a 3-core bus cable, GENIbus

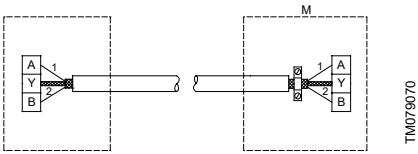
For the bus connection, use a screened 3-core cable with a cross-sectional area of minimum 0.5 mm^2 and maximum 1.5 mm^2.

- If the motor is connected to a unit with a cable clamp identical to the one on the product, connect the screen to the cable clamp.
- If the unit has no cable clamp, leave the screen unconnected at this end.



5.7.2 Replacing a motor

If a 2-core cable is used in the installation, connect it as shown in the figure below.



Connection with screened 2-core cable

Pos.	Description
M	Motor

If a screened 3-core cable is used in the installation, follow the instructions in the section on connecting a 3-core bus cable.

Related information

5.7.1 Connecting a 3-core bus cable, GENIbus

5.8 Signal relays

The motor has two outputs for potential-free signals via two internal relays. You can set the signal outputs to **Operation**, **Pump running**, **Ready**, **Alarm** and **Warning**.

The functions of the two signal relays are shown in the table below.

If you want to set the Pt100/1000 input for a feedback sensor, we recommend that you do this via the **Assisted pump setup** menu. See the section on assisted pump setup.

If you want to set a Pt100/1000 input for other purposes, you can do this manually.

You can set the analog inputs via the **Setup, analog input** menu. See the section on setup, analog input.

If you make the manual setting via the Grundfos GO, you need to enter the menu for the Pt100/1000 input under the **Settings** menu.

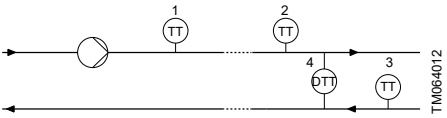
Function

The Pt100/1000 inputs can be set to the following functions:

- **Not active**
- **Feedback sensor**: The sensor is used for the selected control mode.
- **Ext. setpoint infl.**: See the section on the external setpoint function.
- **Other function**.

Measured parameter

Select one of the parameters listed to be measured in the system by the PT100/1000 sensor connected to the actual PT100/1000 input.



Overview of PT100/1000 sensor locations

Parameter	Pos.
Liquid temp.	1
Temperature 1	2
Temperature 2	3
Ambient temp.	Not shown

Measuring range

-50 to 204 °C.

Related information

6.8 Identification of the functional module
7.16 External setpoint function
7.46 Assisted pump setup
7.47 Setup, analog input
7.52 Factory settings

7.10 Digital inputs

Available inputs depend on the functional module fitted in the pump:

Function (terminal)	FM 200 [8] (standard)	FM 300 [8] (advanced)
Digital input 1, setup (2 and 6)	•	•
Digital input 2, setup (1 and 9)	-	•

8) See the section on the identification of the functional module.

To set a digital input, make the settings below.

Function

Select one of the following functions:

- **Not active**: When set to **Not active**, the input has no function.
- **External stop**: When the input is deactivated (open circuit), the pump stops.
- **Min.** (minimum speed): When the input is activated, the pump runs at the set minimum speed.
- **Max.** (maximum speed): When the input is activated, the pump runs at the set maximum speed.
- **User-defined speed**: When the input is activated, the motor runs at a speed set by the user.
- **External fault**: When the input is activated, a timer is started. If the input is activated for more than 5 seconds, the pump stops and a fault is indicated. This function depends on input from external equipment.
- **Alarm resetting**: When the input is activated, a possible fault indication is reset.
- **Dry running**: When this function is selected, lack of inlet pressure or water shortage (dry running) can be detected, and if that occurs, the pump stops. The pump cannot restart as long as the input is activated. This requires the use of an accessory, such as the following:
  - a pressure switch installed on the inlet side of the pump
  - a float switch installed on the inlet side of the pump.
- **Accumulated flow**: When this function is selected, the accumulated flow can be registered. This requires the use of a flowmeter that can give a feedback signal as a pulse per defined volume of water. See the section on pulse flowmeter setup.
- **Predefined setpoint digit 1** (applies only to digital input 2): When digital inputs are set to predefined setpoint, the pump operates according to a setpoint based on the combination of the activated digital inputs. See the section on predefined setpoints.

- **Active output**: When the input is activated, the related digital output is activated. See the section on digital inputs/outputs. This is done without any changes to pump operation.
- **Local motor stop**: When the input is activated, the given pump in a multipump system stops without affecting the performance of the other pumps in the system.

The priority of the selected functions in relation to each other is listed in the section on priority of settings.

A stop command always has the highest priority.

Activation delay

Select the activation delay (T1).

It is the time between the digital signal and the activation of the selected function.

Range: 0-6000 seconds.

Duration timer mode

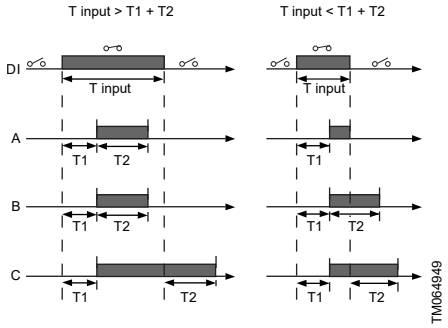
Select the mode. See the figure below.

- **Not active**
- active with interrupt (mode A)
- active without interrupt (mode B)
- active with after-run (mode C).

Select the duration time (T2).

It is the time that, together with the mode, determines how long the selected function is active.

Range: 0 to 15,000 seconds.



Duration timer function of digital inputs

Pos.	Description
DI	Digital input
A	Mode A
B	Mode B
C	Mode C

Related information

6.8 Identification of the functional module
7.1 Priority of settings
7.11 Digital inputs/outputs
7.17.1 Predefined setpoints
7.23 Pulse flowmeter setup
7.52 Factory settings

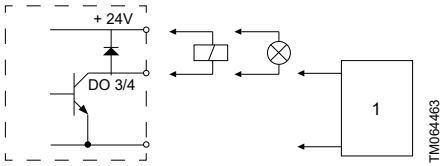
7.11 Digital inputs/outputs

Available inputs/outputs depend on the functional module fitted in the pump:

Function (terminal)	FM 200 [9] (standard)	FM 300 [9] (advanced)
Digital input/output 3, setup (6 and 10)	•	•
Digital input/output 4, setup (11 and 18)	-	•

9) See the section on the identification of the functional module.

You can select if the interface is to be used as input or output. The output is an open collector and you can connect it to, for example, an external relay or controller such as a PLC.



Example of configurable digital inputs/outputs

Pos.	Description
1	External controller

To set a digital input/output, make the settings below.

Mode

You can set the digital input/output 3 and 4 to act as digital input or digital output:

- **Digital input**
- **Digital output**.

Function

You can set the digital input/output 3 and 4 to the functions stated in the table below:

When you enable the LiqTec function, it stops the pump if dry running occurs. Restart the pump manually if it stops due to dry running.

Dry running detection delay

You can set a detection delay to make sure the pump is given a chance to start up before the LiqTec function stops it due to dry running.

Range: 0-254 seconds.

Related information

7.52 Factory settings

7.20 Stop function (Low-flow stop function)

You can set **Low-flow stop function** to the following values:

- **Not active**
- **Energy-optimal mode**
- **High-comfort mode**
- **User-defined mode** (**Customised operating mode**).

When the low-flow stop function is active, the flow is monitored. If the flow becomes lower than the set minimum flow (Q_{min}), the pump changes from continuous operation at constant pressure to start-stop operation and stops if the flow reaches zero.

The advantages of enabling **Low-flow stop function** are the following:

- There is no unnecessary heating from the pumped liquid.
- The wear of the shaft seals is reduced.
- There is reduced noise from operation.

The disadvantages of enabling **Low-flow stop function** may be the following:

- The delivered pressure is not completely constant as it fluctuates between the start and stop pressures.
- The frequent starts and stops of the pump may in some applications cause acoustic noise.

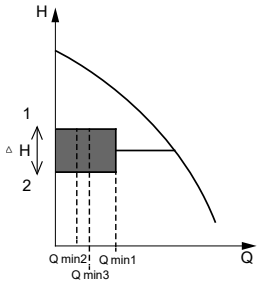
The impact of the above disadvantages very much depends on the setting selected for the stop function.

The **High-comfort mode** setting minimises pressure fluctuations and acoustic noise.

Select **Energy-optimal mode** if the main priority is to reduce the energy consumption as much as possible.

Possible settings of the stop function:

- **Energy-optimal mode**: The pump automatically adjusts the parameters for the stop function so that the energy consumption during the start-stop operation period is minimised. In this case, the stop function uses the factory-set values of the minimum flow (Q_{min1}) and other internal parameters. See the figure below.
- **High-comfort mode**: The pump automatically adjusts the parameters for the stop function so that the disturbances during the start-stop operation period are minimised. In this case, the stop function uses the factory-set values of the minimum flow (Q_{min2}) and other internal parameters. See the figure below.
- **User-defined mode** (**Customised operating mode**): The pump uses the parameters set for ΔH and minimum flow (Q_{min3}) respectively for the stop function. See the figure below.



Difference between start and stop pressures (ΔH) and minimum flow rate

Pos.	Description
1	Stop pressure
2	Start pressure

In start-stop operation, the pressure varies between the start and stop pressures. See the figure above.

In **User-defined mode** (**Customised operating mode**), ΔH is factory-set to 10 % of the actual setpoint. ΔH can be set within the range from 5 to 30 % of the actual setpoint.

The pump changes to start-stop operation if the flow becomes lower than the minimum flow rate.

The minimum flow rate is set in percentages of the nominal flow rate of the pump. See the pump nameplate.

In **User-defined mode** (**Customised operating mode**), the minimum flow rate is factory-set to 10 % of the nominal flow rate.

Low-flow stop function

A low flow can be detected in two ways:

1. One possibility is a built-in low-flow detection function that is active if none of the digital inputs are set for flow switch.
   - Low-flow detection function: The pump checks the flow regularly by reducing the speed for a short time. If there is no or only a small change in pressure, this means that there is low flow. The speed is increased until the stop pressure (actual setpoint + 0.5 × ΔH) is reached and

the pump stops. When the pressure falls to the start pressure (actual setpoint - 0.5 × ΔH), the pump restarts.

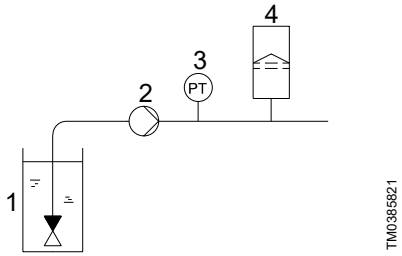
- If the flow rate is higher than the set minimum flow rate, the pump returns to continuous operation at constant pressure.
- If the flow rate is still lower than the set minimum flow rate (Q_{min}), the pump continues in start-stop operation until the flow rate is higher than the set minimum flow rate (Q_{min}). When the flow rate is higher than the set minimum flow rate (Q_{min}), the pump returns to continuous operation.

2. The other possibility is a flow switch connected to one of the digital inputs.
   - Flow switch: When the digital input is activated for more than 5 seconds due to low flow, the speed is increased until the stop pressure (actual setpoint + 0.5 × ΔH) is reached, and the pump stops. When the pressure falls to start pressure, the pump restarts. If there is still no flow, the pump quickly reaches the stop pressure and stops. If there is flow, the pump continues operating according to the setpoint.

Operating conditions for the low-flow stop function

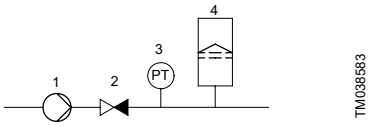
You can only use the stop function if the system incorporates a pressure sensor, a non-return valve and a diaphragm tank.

Always install the non-return valve before the pressure sensor and diaphragm tank.



Position of the non-return valve and pressure sensor in system with suction lift operation

Pos.	Description
1	Non-return valve
2	Pump
3	Pressure sensor
4	Diaphragm tank



Position of the non-return valve and pressure sensor in a system with a positive inlet pressure

Pos.	Description
1	Pump
2	Non-return valve
3	Pressure sensor
4	Diaphragm tank

Set minimum flow

Set the minimum flow rate (Q_{min}) in this display.
This setting determines at which flow rate the system is to change from continuous operation at constant pressure to start-stop operation. The setting range is 5 to 30 % of the rated flow rate.

Related information

7.52 Factory settings

7.21 Stop at min. speed

This stop function can be utilised in, for example, constant level applications where a boost of pressure is not needed. This stop function differs from the low-flow stop but the purpose is the same. The pump stops if there is no or low consumption.

This function monitors the speed of the pump. When the PI-controller has forced the speed of the pump to minimum according to the feedback value, the pump stops after a set period of time. It remains stopped until the feedback value starts to drop and the PI-controller starts the pump again.

- **Enable Stop at min. speed**
  Enables the function **Stop at min. speed**.
- **Delay**
  The delay time during which the pump must be running at minimum speed before it stops.
- **Restart speed**
  It is the speed in percentage when the pump must start again, hysteresis. It must be set higher than the minimum speed of the pump.

7.22 Pipe filling function

This function is typically used in pressure-boosting applications and ensures smooth startup of systems with, for instance, empty pipes.

4. Read the description for this control mode.
5. Select which analog input to use as sensor input.
6. Select the sensor function according to where the sensor is installed in the system. See the section on analog inputs.
7. Select the electrical input signal according to the sensor specifications.
8. Select the measuring unit according to the sensor specifications.
9. Set the minimum and maximum sensor range values according to the sensor specifications.
10. Set the desired setpoint.
11. Set the gain and integral time of the controller. See the section on the controller.
12. Type the desired pump name.
13. Check the summary of settings and confirm them.

On the advanced operating panel

1. Open the **Assist** menu.
2. Select **Assisted pump setup**.
3. Select the control mode **Const. pressure**.
4. Select which analog input to be used as sensor input.
5. Select the measured parameter which is to be controlled. See the section on analog inputs.
6. Select the measuring unit according to the sensor specifications.
7. Set the minimum and maximum sensor range values according to the sensor specifications.
8. Select electrical input signal according to the sensor specifications.
9. Set the desired setpoint.
10. Set the gain and integral time of the controller. See the section on the controller.
11. Check the summary of settings and confirm them by pressing [OK].

Related information

7.2 Setpoint
7.6 Control mode
7.8 Analog inputs
7.14 Controller settings

7.47 Setup, analog input

This menu is only available in the advanced operating panel.

This menu guides you through the following:

Setup, analog input

- Analog inputs 1 to 3.
- Pt100/1000 input 1 and 2.
- Adjusting the setpoint.
- Summary.

Related information

7.2 Setpoint
7.8 Analog inputs
7.9 Pt100/1000 inputs

7.48 Setting of date and time

This function is only available in the HMI 300 and HMI 301 operating panels.

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Setting of date and time
FM110	-
FM310	●
FM311	●

The function guides you through the following settings:

- **Select date format**
- **Set date**
- **Select time format**
- **Set time**.

Related information

6.8 Identification of the functional module
7.32 Set date and time

7.49 Setup of multipump system

The function **Setup of multi-pump system** enables the control of two motors connected in parallel without external controllers. The pumps or motors in a system communicate with each other via the wireless GENIair connection or the wired GENI connection.

You can set a multipump system via the master motor which is the first selected motor.

If several pumps or motors in the system have sensors, they can all function as the master and take over the master function if the other fails. This provides additional redundancy in the multimotor system.

You can choose between the following multimotor functions:

Alternating operation

Alternating operation serves as a duty and standby operating mode, and is possible with two pumps or two motors of the same size and type connected in parallel. The main purpose of the function is to ensure an even amount of running hours, and to ensure that the other pump or motor starts if the duty pump or motor stops due to an alarm.

You can choose between two alternating operating modes:

- **Alternating operation, time**:

  The change from one pump or motor to the other is based on time.
- **Alternating operation, energy**:

  The change from one pump or motor to the other is based on energy consumption.

If the duty pump or motor fails, the other pump or motor starts.

Backup operation

Backup operation is possible with two motors of the same size and type connected in parallel. One motor is operating continuously. The backup motor is operated for a short time each day to prevent seizing up. If the duty motor stops due to a fault, the backup motor starts.

Cascade operation

This function is available with up to 4 motors installed in parallel. The motors must be of the same size and if used with pumps, the pumps must be of the same model.

- The performance is adjusted to the demand through cutting pumps in or out and through parallel control of the pumps in operation.
- The controller maintains a constant process value through continuous adjustment of the speed of the pumps.
- Pump changeover is automatic and depends on load, operating hours and fault detection.
- All pumps in operation run at the same speed.
- The number of pumps in operation also depends on the energy consumption of the pumps. If only one pump is required, two pumps run at a lower speed if this results in a lower energy consumption.
- If several pumps or motors in the system have a sensor, they can all function as the master and take over the master function if the other fails.

7.49.1 Alternating operation, time

The **Alternating operation, time** menu sets the interval of alternation between two pumps.

This setting is only available in alternating mode.

7.49.2 Time for pump changeover

The **Time for pump changeover** menu sets the time of day for pump changeover to take place.

This setting is only available in alternating operation.

7.49.3 Sensor to be used

This function defines the sensor for controlling the pump system.

Select **Master pump sensor** if the sensor is placed in a way where it can measure the output from all the pumps in the system, for example, in the manifold.

Select **Running pump sensor** if the sensor is placed on or across individual pumps, for example, if the sensor is installed behind non-return valves, or if it is not able to measure the output from all pumps.

This setting is only available in alternating operation and cascade operation.

7.49.4 Setting up a multipump system

You can set up a multipump system in the following ways:

- Grundfos GO and wireless motor connection
- Grundfos GO and wired motor connection
- HMI 300 or HMI 301 operating panel and wireless motor connection
- HMI 300 or HMI 301 operating panel and wired motor connection.

Related information

7.29 Pump number

7.40 Pump name

7.46 Assisted pump setup

7.49.4.1 Grundfos GO and wireless motor connection

1. Power on both motors.
2. Establish contact to one of the motors with the Grundfos GO.
3. Set the needed analog and digital inputs via the Grundfos GO, according to the connected equipment and the required functionality.
4. Assign a name to the motor using the Grundfos GO.
5. Disconnect the Grundfos GO from the motor.
6. Establish contact to the other motor.
7. Set the needed analog and digital inputs via the Grundfos GO, according to the connected equipment and the required functionality.
8. Assign a name to the motor using the Grundfos GO.
9. Select the **Assist** menu and **Setup of multi-pump system**.
10. Select the desired multimotor function.
11. Press the **Right** button to continue.
12. Set the time when the alternation between the two motors is to take place.

    This step applies only if you have selected **Alternating operation, time** and if the motors are fitted with the FM310 or FM311.
13. Press the **Right** button to continue.

14. Select **Radio** as the communication method between the two motors.
15. Press the **Right** button to continue.
16. Select pump 2 (motor 2).
17. Select the pump from the list.



Use **OK** or the **Radio communication** button to identify the pump.

18. Press the **Right** button to continue.
19. Confirm the setting by pressing **Send**.
20. When you finish the setup and the dialog box disappears, wait for the green indicator light in the middle of **Grundfos Eye** to light up.

7.49.4.2 Grundfos GO and wired motor connection

1. Connect the two motors with each other with a 3-core screened cable between the GENIbus terminals A, Y, B.
2. Power on both motors.
3. Establish contact to one of the motors with the Grundfos GO.
4. Set the required analog and digital inputs via the Grundfos GO, according to the connected equipment and the required functionality.
5. Assign a name to the motor using the Grundfos GO.
6. Assign motor number 1 to the motor.
7. Disconnect the Grundfos GO from the motor.
8. Establish contact to the other motor.
9. Set the analog and digital inputs according to the connected equipment and the required functionality by the Grundfos GO.
10. Assign a name to the motor using the Grundfos GO.
11. Assign motor number 2 to the motor.
12. Select the **Assist** menu and **Setup of multi-pump system** (**multimotor setup**).
13. Select the desired multimotor function.
14. Press the **Right** button to continue.
15. Set the time when the alternation between the two motors is to take place.



This step applies only if you have selected **Alternating operation, time** and if the motors are fitted with the FM310 or FM311.

16. Press the **Right** button to continue.
17. Select **Bus** as the communication method between the two motors.
18. Press the **Right** button to continue.
19. Select pump 2 (motor 2).
20. Select the additional motor from the list.



Use **OK** or the **Radio communication** button to identify the pump.

21. Press the **Right** button to continue.
22. Confirm the setting by pressing **Send**.
23. When you have finish the setup and the dialog box disappears, wait for the green indicator light in the middle of the **Grundfos Eye** to light up.

7.49.4.3 HMI 300 or HMI 301 and wireless motor connection

1. Power on both motors.
2. On both motors, set the analog and digital inputs according to the connected equipment and the required functionality.
3. Select the **Assist** menu on one of the motors and **Setup of multi-pump system**.
4. Press the **Right** button to continue.
5. Select **Wireless** as the communication method between the two motors.
6. Press the **Right** button to continue.
7. Select the desired multimotor function.
8. Press the **Right** button three times to continue.
9. Press **OK** to search for other motors.

   The green indicator light in the middle of the **Grundfos Eye** flashes on the other motors.
10. Press **OK** or the **Radio communication** button on the motor that is to be added to the multimotor system.
11. Press the **Right** button to continue.